

X_{comp} : definitions, dependencies, and the physics behind it

Compiled for the `photon_vs_electron` project

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1 What X_{comp} is

At luminosity parity $L_\gamma = L_e$, X_{comp} compares how many ionizations each agent buys with a given amount of injected energy:

$$X_{\text{comp}}(\alpha_\gamma, \alpha_e) = \frac{\int S_\gamma N_{\text{ion},\gamma} dE / \langle E \rangle_\gamma}{\int S_e N_{\text{ion},e} dE / \langle E \rangle_e} = \frac{\langle N_\gamma \rangle / \langle E \rangle_\gamma}{\langle N_e \rangle / \langle E \rangle_e} = \frac{W_e(\alpha_e)}{W_\gamma(\alpha_\gamma)}, \quad (1)$$

where $W \equiv \langle E \rangle / \langle N_{\text{ion}} \rangle$ is the *energy price of one ionization* — the W -value of radiation physics. $X_{\text{comp}} > 1$ means photons are the cheaper ionizing agent per erg.

Everything else cancels. n_{H} , the comoving conversion, the redshift and the luminosity normalisation all drop out of eq. (1) identically, so the map depends on α_γ and α_e *and nothing else*. Because $\log X_{\text{comp}} = \log W_e(\alpha_e) - \log W_\gamma(\alpha_\gamma)$ is additively separable, the map has no degenerate direction: both axes do real work.

2 Two repairs to the construction as originally written

(1) The denominator double-counted the yield. The integrand was specified as $S(E)N_{\text{ion}}(E)/\langle E_{\text{ion}} \rangle$ with $\langle E_{\text{ion}} \rangle$ defined as “the average energy spent per ionization”, i.e. W itself. With S number-normalised that gives $\langle N \rangle / W = \langle N \rangle^2 / \langle E \rangle$: the yield enters twice. The form is *dimensionally* correct either way, which is why it does not announce itself — but the two readings differ by $\langle N_\gamma \rangle / \langle N_e \rangle \approx 1/106$ and **invert the conclusion**: $X_{\text{comp}} = 2.07$ (photons win) versus 0.0196 (electrons win 51 $\times$). Adopted here: $\langle E_{\text{ion}} \rangle = \langle E \rangle$, the mean energy per *primary*, which makes the integral the ionizations produced per unit energy injected.

(2) The band top is derived, not adopted. “The free-streaming regime ($\sim$ keV)” evaluates, using this project’s own σ_{photoion} over one Hubble length at $z = 10$, to $\tau_{\text{IGM}} = 1$ at $E = 1218.0$ eV. A smooth $(1 - e^{-\tau})$ weighting was tried first and *diverges*: it charges the numerator $\langle E \rangle$ for photons that free-stream and ionize nothing while the denominator saturates, so W_γ grows without bound (measured +224% at $\alpha_\gamma = 0.5$ between a 10 keV and a 100 keV integration limit). The sharp band is therefore kept.

3 The physics: why electrons are the more expensive agent

The floor. Ionizing hydrogen costs $E_{\text{th}} = 13.598$ eV and no less, so E_{th}/W is a true efficiency. At the fiducial indices ($\alpha_\gamma = 2.0$, $\alpha_e = 2.2$):

	W [eV/ion]	efficiency E_{th}/W	$\langle N_{\text{ion}} \rangle$
photons	21.075	64.5%	1.0325
electrons	48.038	28.3%	122.92

so $X_{\text{comp}} = 2.2794$: a photon buys an ionization for less than half the energy.

Where the electron’s energy actually goes. Averaged over the injection spectrum, the loss flux divides as:

mechanism	fraction of dK/dt
ionization	72.07%
excitation	26.56%
compton	1.08%
adiabatic	0.22%
coulomb	0.07%

The decisive entry is **excitation at 26.56%**. That energy leaves as Lyman-series photons, *every one of which is below 13.6 eV* — Ly α is 10.2 eV — so it can never ionize hydrogen, at any later time, by any route. It is not delayed; it is gone. A further 1.37% goes to inverse Compton on a CMB that is $(1+z)^4 \simeq 1.5 \times 10^4$ times denser at $z = 10$, plus adiabatic and Coulomb losses. The remainder is sub-threshold kinetic energy: once a secondary electron drops below 13.6 eV it can only excite and heat, so it too leaves the ionization budget. Of the energy injected, only 28.3% ends up as binding energy delivered.

Why photons escape most of that penalty. A photon does not avoid the electron physics — it *defers* it. Photoionization converts the first 13.598 eV at 100% efficiency, and only the excess $E - E_{\text{th}}$ is handed to a photoelectron, which from that instant suffers exactly the same losses as any CR electron. Structurally,

$$N_{\text{ion},\gamma}(E) \simeq 1 + N_{\text{ion},e}(E - E_{\text{th}}), \quad (2)$$

one free ionization plus an electron cascade. Checked directly:

E [eV]	$N_{\text{ion},\gamma}(E)$	$1 + N_{\text{ion},e}(E - E_{\text{th}})$	rel. diff.
54.4	1.858	1.724	7.8%
100	2.963	3.077	3.7%
1000	26.364	28.977	9.0%
10^4	265.337	277.162	4.3%
10^5	2157.676	2203.241	2.1%

Equation (2) is *approximate*, not an identity: the few-per-cent residual is the known disagreement between the depfit-route photon route and the loss+IC-route electron route (assumption A9, 7.7%), not new physics.

The consequence. The photon advantage is not that photons are intrinsically efficient — it is that they can be delivered *at the threshold, where the job is cheapest*, whereas CR electrons are injected at keV–TeV and must bleed down through non-ionizing channels to get there. Monoenergetically the two costs converge above ~ 1 keV (37.9 vs 35.3 eV/ion at 1 keV; 46.3 vs 45.4 at 100 keV), because a keV photon’s photoelectron *is* a keV electron. The entire photon advantage lives in the 13.6–100 eV window, and is really a statement about how badly slow electrons perform there: a 20 eV photon ionizes with near-certainty, while a 20 eV electron costs 171 eV per ionization because it overwhelmingly excites instead.

4 The weakest input: $E_{\text{min},e}$

The two bands are not equally well determined. The photon band’s floor is atomic physics and its ceiling is derived from $\tau = 1$; *every* entry of the electron band is a choice with no external source. That would be harmless if the answer were insensitive to it:

$E_{\min,e}$ [eV]	W_e [eV/ion]	X_{comp} at fiducial indices
10^2	43.066	1.8585
10^3	48.038	2.0730
10^4	60.929	2.6293
10^5	94.470	4.0768

Lowering $E_{\min,e}$ drags in the catastrophically expensive sub-100 eV region; raising it does the reverse. Across 10^2 – 10^5 eV the whole map slides by a factor 2.19 — **more than the entire α_e axis moves it**. The map’s *shape* is well posed; its *normalisation* is not. The figure is therefore drawn at two values of $E_{\min,e}$ rather than one, so the shift is visible rather than hidden in a constant.

5 Inventory

Status: **V** verified against a source in **papers/**; **C** cited, unverified; **D** derived here; **S** scanned; **U** your choice; **X** unsourced.

symbol	value	units	depends on	st.
Scenario				
z redshift of the snapshot [src: project fiducial]	10	–	–	U
n_{H} proper hydrogen number density [src: Planck2020]	2.5287×10^{-4}	cm^{-3}	$z, \Omega_b h^2, X_{\text{H}}$	D
x_e ionized fraction, held static [src: your value]	10^{-4}	–	–	U
$H(z), t_{\text{H}}$ expansion rate and Hubble time [src: Planck2020]	0.7086 Gyr	$\text{s}^{-1}, \text{Gyr}$	Planck18 complete $H(z)$	D
target pure atomic hydrogen at mean density [src: project fiducial]	$Z = 0$	–	–	U
Photon band				
$E_{\min,\gamma}$ HI ionization potential – PHYSICAL floor [src: NIST_ASD]	13.598434600	eV	atomic constant	V
$E_{\text{max,g}}$ (stellar) band top ACTUALLY USED: min(4 Ryd source cutoff, IGM cutoff) [src: DERIVED here]	54.4228	eV	band policy	D
$E_{\text{max,g}}$ (IGM) free-streaming onset: $\tau_{\text{IGM}} = 1$ over one Hubble length; caps any source whose own emission reaches past it [src: DERIVED here]	1218.0	eV	$\sigma_{\text{photoion}}, n_{\text{HI}}, c/H(z)$	D
$\tau_{\text{IGM}}(E)$ IGM optical depth over a Hubble length [src: Karzas1961 / DERIVED]	1.934 at 1 keV, 0.0476 at 3 keV	–	$\sigma_{\text{photoion}}, n_{\text{HI}}, z$	D
$S_{\gamma}(E)$ $dN/dE \propto E^{-(\alpha_{\gamma}+1)}$, NUMBER-normalised to 1 on the band [src: convention]	power law	eV^{-1}	$\alpha_{\gamma}, \text{band}$	U
α_{γ} (fiducial) the index actually used, $f_{\nu} \propto \nu^{-\alpha_{\gamma}}$. OUR PARAMETRISATION: MarquesChaves2026 gives the population (BPASS v2.2.1, $Z=0.003$, 6.8 Myr) but no LyC index, and its beta_UV is measured longward of 1400 Å, which cannot be carried across the Lyman break [src: this work]	2.0	–	–	U
α_{γ} (measured) the ONLY direct measurement in our band: $L_{\text{nu}} = \int \nu^{\alpha_{\gamma}} d\nu$ over 13.6–54.4 eV, exactly this project’s convention, no conversion. CAVEAT: four GALACTIC HII regions at roughly solar metallicity, and the authors find it disagrees with BPASS; metal-poor $z=10$ should be harder, i.e. smaller [src: Murchikova2020]	4.5 ± 0.4	–	–	V
α_{γ} (axis)	0.3–5.0	–	–	S

symbol	value	units	depends on	st.
range swept by the figure, chosen to contain every index sourced in this project AND the measured 4.5 [src:]				
Electron band				
$E_{\min,e}$	10^3	eV	–	U
low cutoff of the CR injection spectrum [src: your choice]				
$E_{\max,e}$	10^{12}	eV	–	U
high cutoff of the CR injection spectrum [src: your choice]				
$S_e(E)$	power law	eV^{-1}	α_e , band	U
$dN/dE \propto E^{-\alpha_e}$, NUMBER-normalised to 1 on the band [src: convention]				
α_e	2.2 (fid); 1.2-2.7 (axis)	–	–	U
CR electron injection index [src: your choice; DSA predicts 2]				
Ionization yields				
$N_{\text{ion},\gamma}(E)$	1.0325 (band mean)	–	E, x_e , z	V
ionizations per ABSORBED photon of energy E, primary + full secondary cascade [src: ionization.yield, depfit route]				
$N_{\text{ion},e}(E)$	122.92 (band mean)	–	E, x_e , z, B	V
ionizations per INJECTED electron of energy E, 7 loss channels + BED cascade + IC secondaries [src: igm.losses the loss+IC route]				
Derived moments				
$\langle E \rangle_\gamma$	21.7598	eV	α_γ , band	D
mean energy per emitted ionizing photon [src: DERIVED]				
$\langle E \rangle_e$	5.9049×10^3	eV	α_e , band	D
mean energy per injected CR electron [src: DERIVED]				
W_γ	21.0747	eV ion^{-1}	α_γ , band, x_e , z	D
energy price of one ionization, photons = $\langle E \rangle_\gamma / \langle N \rangle_\gamma$ [src: DERIVED]				
W_e	48.0378	eV ion^{-1}	α_e , band, x_e , z, B	D
energy price of one ionization, electrons = $\langle E \rangle_e / \langle N \rangle_e$ [src: DERIVED]				
X_{comp}	2.2794	–	α_γ , α_e ONLY	D
W_e/W_γ at $L_\gamma = L_e$: ionizations per unit energy, photons relative to electrons [src: DERIVED]				